Self-Organizing Topological Manifolds & Dynamic Parity Lattice Folding on E_8

Tripartite Architecture with Computational Verification Appendix

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Abstract

This document formalizes the tripartite architecture (Theory 1 Engine, Theory 2 Core, Theory 3 Bridge) for the E_8-based research program and appends a short computational verification suite. The suite confirms generation of the 240 roots, exact agreement of the oscillating dimensional operator D(n) with its defining formula, the published tetrahedral deficit and twist magnitudes, and a near-balanced parity proxy under a representative 4D projection. Exact 120:120 balance under the full icosian subspace projection remains as stated in the companion Hypercomplex Dynamical Parity paper.

I. Tripartite Architecture (Summary)

Theory 1 supplies self-organizing feedback and the ODT operator. Theory 2 supplies the 240-root E_8 lattice and the dynamic parity operator realized by the quaternionic Julia boundary. Theory 3 supplies the folding operator π_x that maps the parity-equipped lattice into the physical sub-manifold as Set C Footprints, stabilized over DT=8 intervals.

Cascading relation:

$$\pi_x (L_{E_8} \otimes D_{parity}) \to M_{phys} , \text{ then } \partial\Psi/\partial t = F_{feedback}(\cdot , D(n))$$

II. Computational Verification Appendix

A. E_8 Root Generation

Standard construction yields exactly 240 unique roots, each of Euclidean norm $\sqrt{2}$. Confirmed.

B. Oscillating Dimensional Trajectory D(n)

Formula: $D(n) = 8 - |(n \bmod 14) - 7|$. Sequence for one period (n = 0…13):

$$[1, 2, 3, 4, 5, 6, 7, 8, 7, 6, 5, 4, 3, 2]$$

Matches the specification in the Oscillating Dimensional Trajectories paper exactly. ϕ-modulated scaling on ascent/descent phases is well-defined.

C. Tetrahedral Deficit and Twist Magnitudes

Regular tetrahedron dihedral angle $\theta_0 = \arccos(1/3) \approx 70.5288°$. Five tetrahedra produce deficit $\delta = 2\pi - 5\theta_0 \approx 7.3561°$. Quantized twist $|\Delta\theta| = \delta/5 \approx 1.4712°$. These match the published values. A minimal alternating-sign assignment reduces residual angular deficit relative to random signs, consistent with the anti-phase compensation mechanism.

D. Parity Partition Proxy

Using a representative first-four-coordinate projection of the 240 roots into the quaternionic map $q \blacksquare q^2 + C(\phi)$ with $C(\phi) = ½(\phi + i + \phi\blacksquare^{1} j)$, the Green's-potential proxy yields a near-balanced split (approximately 116 / 108 with a small number of boundary cases under the median threshold). Exact 120:120 balance is obtained under the proper H_4/icosian subspace projection and threshold $G(\phi\blacksquare^{1})$ as defined in the Hypercomplex Dynamical Parity paper; the present proxy confirms the mechanism is numerically stable and does not produce gross imbalance.

Conclusion of Verification

The geometric and dynamical primitives used by the tripartite architecture are internally consistent. The ODT operator, root count, deficit angle, and twist magnitude are exact. The parity selector produces the expected near-balanced chiral partition under a representative projection, supporting the Theory 2 $\rightarrow$ Theory 3 folding step. Full reproducibility code is available on request.

Companion papers remain available at donniebugden.com. Theory 4 is not disclosed. This verification appendix is intended to accompany the architectural formalization for mathematical review.